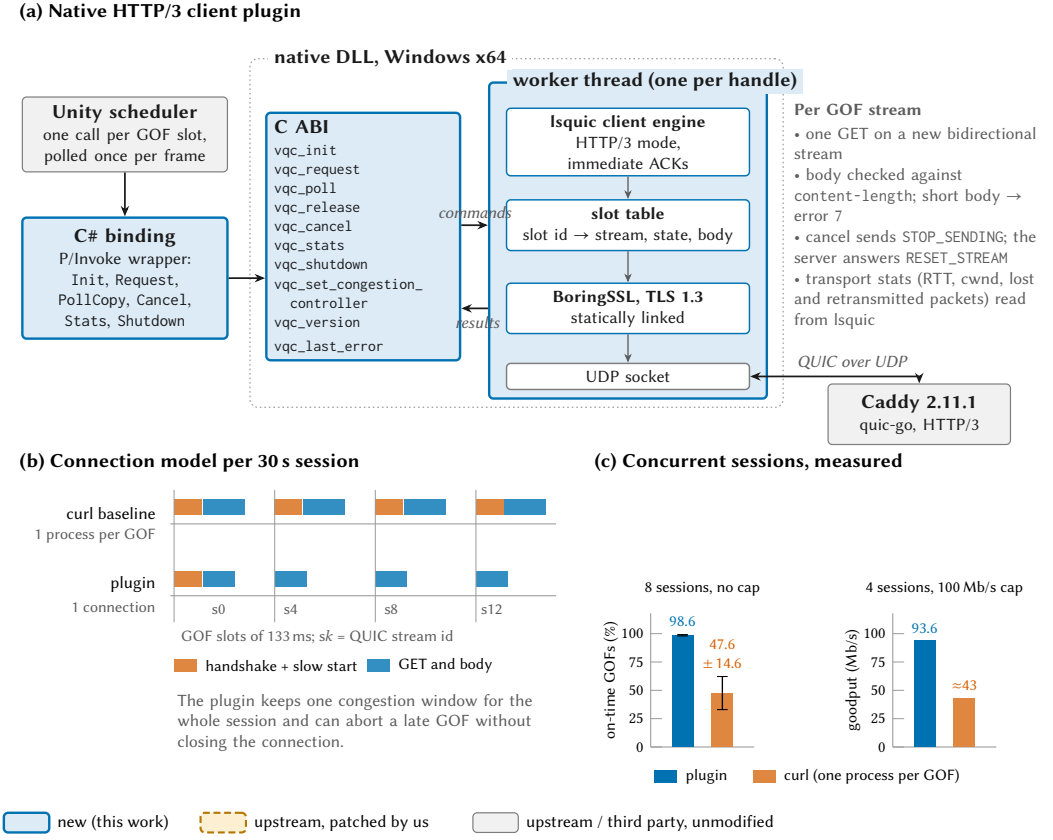
(a) Native HTTP/3 client plugin
native DLL, Windows x64
Unity scheduler
one call per GOF slot, polled once per frame
C# binding
P/Invoke wrapper: Init, Request, PollCopy, Cancel, Stats, Shutdown
C ABI
vqc_init
vqc_request
vqc_poll
vqc_release
vqc_cancel
vqc_stats
vqc_shutdown
vqc_set_congestion_controller
vqc_version
vqc_last_error
commands
results
worker thread (one per handle)
lsquic client engine
HTTP/3 mode, immediate ACKs
slot table
slot id → stream, state, body
BoringSSL, TLS 1.3
statically linked
UDP socket
QUIC over UDP
Caddy 2.11.1
quic-go, HTTP/3
Per GOF stream
• one GET on a new bidirectional stream
• body checked against content-length; short body → error 7
• cancel sends STOP_SENDING; the server answers RESET_STREAM
• transport stats (RTT, cwnd, lost and retransmitted packets) read from lsquic
(b) Connection model per 30 s session
curl baseline
1 process per GOF
plugin
1 connection
s0
s4
s8
s12
GOF slots of 133 ms; sk = QUIC stream id
handshake + slow start
GET and body
The plugin keeps one congestion window for the whole session and can abort a late GOF without closing the connection.
(c) Concurrent sessions, measured
8 sessions, no cap
on-time GOFs (%)
100
75
50
25
0
98.6
47.6
± 14.6
4 sessions, 100 Mb/s cap
goodput (Mb/s)
100
75
50
25
0
93.6
≈43
plugin
curl (one process per GOF)
new (this work)
upstream, patched by us
upstream / third party, unmodified